SIDDHARTHA SANAPALA

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PROFESSIONAL SUMMARY

- DevOps-focused Software Engineer with hands-on experience designing cloud-native infrastructure, CI/CD automation, Kubernetes deployments, Infrastructure as Code and observability solutions across GCP, AWS and Azure environments.
- Strong backend engineering experience in Python, Django and REST APIs with practical exposure to production deployments, DevSecOps integration and cloud operations.
- Passionate about building reliable, scalable and automated platforms following SRE principles and modern cloud-native practices.

TECHNICAL SKILLS

Programming	:	Python, Java
Backend	:	Django, Django REST Framework , REST APIs, JWT Authentication
Containers & Orchestration	:	Docker, Kubernetes, Helm
Infrastructure as Code	:	Terraform, Ansible
CI/CD	:	Azure DevOps Pipelines, GitHub Actions, Jenkins
Cloud	:	GCP, AWS, Azure(Web Apps)
Observability	:	Prometheus, Grafana, Sentry
DevSecOps	:	Trivy, Snyk, OWASP ZAP, SAST, DAST
Databases	:	PostgreSQL, MySQL, MongoDB, Redis
Tools	:	Git, Postman, Swagger, Linux, Render, Supabase, Vercel

EXPERIENCE

Trainee Software Engineer | Sails Software

Tech Stack: AWS, Azure Web Apps, DevSecOps, Prometheus, Grafana

- Implemented cloud-native deployment workflows across AWS and Azure environments, supporting application hosting, infrastructure management, and operational reliability.
- Built monitoring dashboards using Prometheus and Grafana by configuring metrics collection, visualization, and alerting to improve system observability.
- Integrated DevSecOps security scanning into CI/CD workflows using Trivy, Snyk and OWASP ZAP, improving deployment quality through automated vulnerability detection.
- Diagnosed deployment failures, application issues and infrastructure incidents across cloud environments by collaborating with development and operations teams.
- Deployed a production-ready full-stack application consisting of React.js frontend, Python backend and PostgreSQL database using Azure Web Apps.
- Configured secure network access by exposing only frontend endpoints publicly while restricting backend and database communication using Azure networking policies.
- Managed application deployment lifecycle including configuration updates, runtime troubleshooting and production maintenance across cloud environments.
- Applied SRE operational practices including monitoring, debugging and incident resolution to improve application reliability.

SRE Intern | Sails Software (Onsite)

Tech Stack : GCP, Docker, Kubernetes, Terraform, Jenkins, Azure DevOps Pipelines, Helm, Ansible, Git

- Containerized applications using Docker and deployed them onto Google Kubernetes Engine (GKE), managing deployments, services, ingress resources, rolling updates, and environment-specific configurations using Helm.
- Automated infrastructure provisioning using Terraform following Infrastructure as Code (IaC) principles, enabling repeatable and consistent cloud resource deployments across environments.
- Designed and refined CI/CD pipelines using Jenkins and Azure DevOps Pipelines by introducing build caching, optimizing deployment workflows, and implementing multi-stage Docker builds to reduce image size and improve deployment efficiency.
- Improved application availability and scalability by configuring Kubernetes LoadBalancer/Ingress resources, implementing Horizontal Pod Autoscaling (HPA), and tuning deployment strategies for varying workloads.
- Strengthened cluster security by applying Kubernetes network policies and secure ingress/egress traffic controls, ensuring controlled communication between application components.

- Collaborated through Git-based workflows to troubleshoot deployment failures, container runtime issues, networking problems, and infrastructure-related incidents during real-world project execution.

Python Developer Intern | Jupitos Technologies LLC

Tech Stack : Python, Django, DRF, JWT, PostgreSQL, Git

- Developed modular REST APIs using Django REST Framework with JWT authentication following secure backend design practices.
- Built serializer, viewset and authentication modules supporting role-based user management and secure API access.
- Completed 10+ backend development assignments covering authentication, CRUD operations, API design and database integration using PostgreSQL.

AWS Intern | Brainovision

Tech Stack : EC2, S3, RDS, VPC, IAM, Elastic Load Balancer, Elastic Container Registry, Auto Scaling Groups, CloudWatch

- Implemented AWS infrastructure components including EC2, IAM, VPC, S3, ELB, Auto Scaling Groups and ECR through practical cloud deployment exercises.
- Explored CloudWatch monitoring, AWS Lambda and CI/CD concepts to understand scalable cloud infrastructure and operational workflows.
- Performed hands-on configuration of networking, identity management and compute resources while following AWS architectural best practices.

PROJECTS

RaktaPraptih – Blood Donation Management System | Github link | Web

Tech Stack : Python, Django, DRF, PostgreSQL, Docker, GitHub Actions, JWT, Render, Supabase, Git

- Engineered a production-ready backend for blood donation coordination using Django and DRF, enabling secure API-driven management of donors and requests. Deployed on Render with automated CI/CD,optimized performance during active usage.
- Built RESTful APIs with DRF, integrating JWT authentication and custom throttling, ensuring 100% secure access and API stability under 100 concurrent users.
- Designed role-based authentication with Django, reducing access control overhead by 30% and supporting multiple user types (donors, recipients, admins).
- Containerized application using Docker, eliminating 95% of environment-related errors.
- Configured CI/CD workflows via GitHub Actions, automating pytest, Bandit security scans, and Render deployment.
- Resolved CI/CD pipeline failures and Render 502 errors by restructuring Gunicorn execution and validating Supabase DATABASE_URL, improving deployment success to 100%.
- Strengthened application reliability by integrating Sentry for real-time error tracking, validating API performance under concurrent workloads using Locust load testing, and automating operational alerts through Twilio for timely notification of critical application failures.

Cloud-Native CI/CD Pipeline & Infrastructure Automation (GCP + Azure DevOps)

Tech Stack: GCP, Terraform, Kubernetes (GKE), Helm, Docker, Azure DevOps

- Designed a multi-repository cloud-native architecture integrating application, infrastructure, and deployment layers.
- Implemented CI/CD pipelines using Azure DevOps to automate build, testing, containerization, and release processes.
- Provisioned scalable infrastructure on GCP using Terraform, including VPC networks, GKE clusters, and artifact registry.
- Defined environment-specific deployment workflows (Dev → Stage → Prod) with approval gates for controlled rollouts.
- Managed Kubernetes deployments using Helm with rolling update strategies and ingress-based routing.
- Secured system architecture by isolating backend services within private subnets and regulating external access.

EDUCATION

JNTU-GV College of Engineering, Vizianagaram — Dec 2021 – May 2025
B.Tech in Information Technology — CGPA: 8.32

Municipal Junior College, Nellore — Jul 2018 – Mar 2020
Intermediate (MPC) — CGPA: 9.85

G.V.M.C High School, Visakhapatnam — Jun 2017 – Mar 2018
SSC — CGPA: 10.0